

Forecasting and structural analysis of hierarchical macroeconomic data

formerly

Forecasting Regional Economic Activity

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Macroeconomic hierarchical data



Figure 1: Germany and its 16 regions.

- Short and at low frequency
- Hierarchical structure
- Example: regional accounts

Tracking regional economic activity

- For regional governments - policy planning, public investment
- For federal government - addressing unwanted heterogeneities

Tracking regional economic activity in (semi) real time

Literature

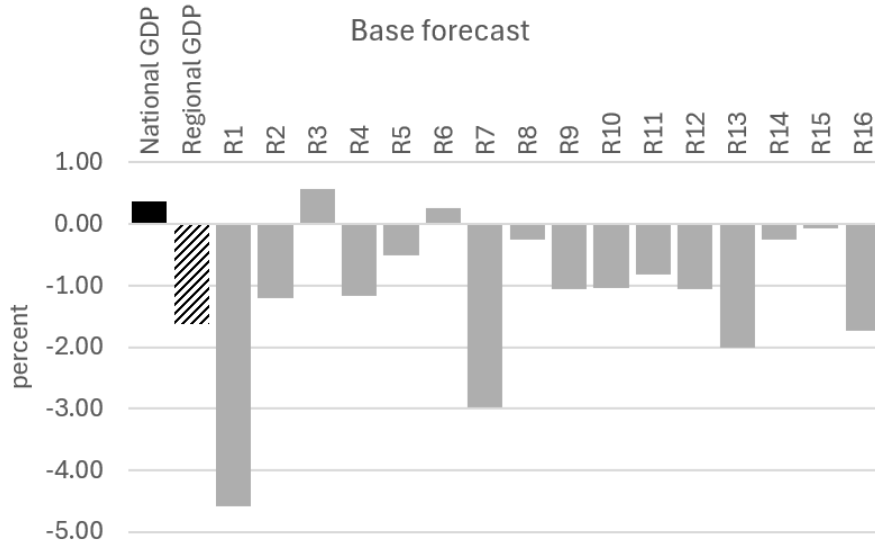
Koop et al., 2020, 2022; Di Fonzo and Girolimetto, 2023; Barbaglia et al., 2024; Koop et al., 2024, 2025; Lehmann and Wikman, 2025

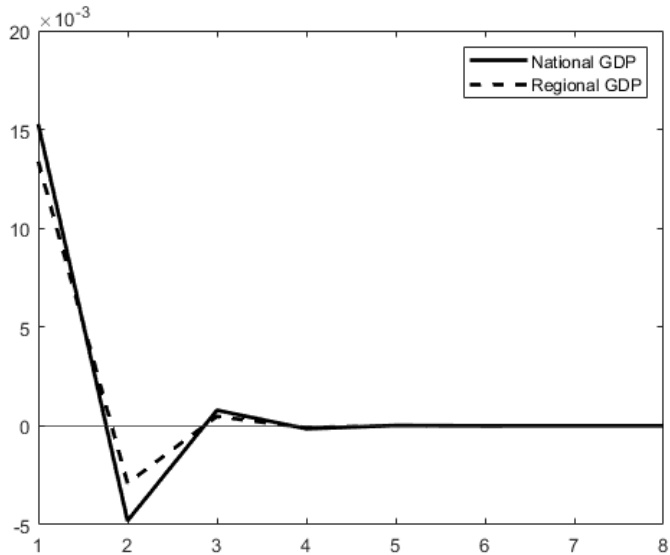
Accurate assessment of regional economic activity

- High-frequency regional data (e.g. quarterly) with great in-sample fit

However...

- Limited to latest federal GDP release (currently Q1 2025)
 - ▶ Governments interested in the full 2025. In autumn - in 2026 for budget planning
- Cannot be used for forecasting and structural analysis
 - ▶ Forecasts and IRFs do not respect the hierarchical structure





The paper in a nutshell

We propose a framework for forecasting and structural analysis of hierarchical macroeconomic data

Forecasting

- Forecast reconciliation:
 - ▶ Model free method for hierarchical forecasting
 - ▶ Unsited for macroeconomic hierarchical data
- ⇒ Alleviate shortcomings through a mixed-frequency framework

Structural analysis

- Reconciled impulse-response functions (rIRF)
 - ▶ IRFs can be calculated through forecasting
 - ▶ these "forecasts" can be reconciled
- ⇒ rIRFs adhere to the hierarchical structure

Forecast reconciliation

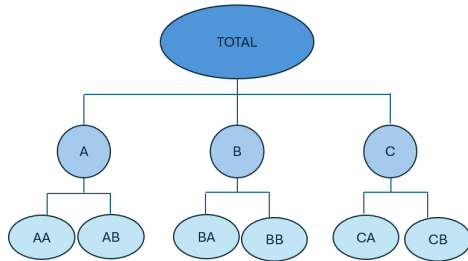


Figure 4: Hierarchical Structure

Method to produce hierarchical forecasts

- 1 “base” forecasts ($\hat{\mathbf{y}}$): incoherent, do not aggregate up
- 2 “reconciled” forecasts ($\tilde{\mathbf{y}}$): respect the hierarchy (regional GDP forecasts add up to federal GDP)

⇒ Forecast reconciliation transforms base forecasts to reconciled forecasts, irrespective of how they were produced

Formulation

Forecast reconciliation is a linear transformation

$$\tilde{\mathbf{y}}_{T+h} = \mathbf{S}\mathbf{P}_h\mathbf{R}\hat{\mathbf{y}}_{T+h}$$

- **S**: *summing* matrix linking regional to national series using a linear combination
- **R**: *selection* matrix indicating which variables in $\hat{\mathbf{y}}$ are hierarchical
- **P_h**: appropriately selected weighting matrix that maps the regional base to the reconciled forecasts

Forecast reconciliation literature

Literature

Wickramasuriya et al., 2019; Eckert et al., 2021; Athanasopoulos et al., 2024; Wickramasuriya, 2024, among others

- The literature concerns with the “optimal” value for $\mathbf{P}$ - in what sense?

$$\min(\tilde{\mathbf{y}}_{T+h} - \mathbf{y}_{T+h}^{\text{true}}) = \min(\mathbf{S}\hat{\mathbf{y}}_{T+h} - \mathbf{y}_{T+h}^{\text{true}})$$

- Solution of (Wickramasuriya et al., 2019)

$$\mathbf{P} = (\mathbf{S}^T \mathbf{W}_h^{-1} \mathbf{S})^{-1} \mathbf{S}^T \mathbf{W}_h^{-1}$$

- $\mathbf{W}_h^{-1} = \text{var}(\hat{\mathbf{y}}_{T+h} - \mathbf{y}_{T+h}^{\text{true}} | \mathcal{I}_T)$ is the base forecast error variance covariance

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Intuition

- base forecast that is close to the “true” value remains relatively unchanged
- base forecast that is far away gets adjusted more
- “true” values respect the cross-sectional constraint (reconciled)

Property

- Reconciled forecasts are *at least as good* and *often better* than base forecasts

Condition

- $\mathbf{W}_h^{-1}$ be “well” estimated (at least $T^{fcst} > N$)
- *Unsuited for regional accounts*

Solution: Mixed-Frequency framework

Chan et al., 2024 derive conditional distribution of *missing* data ($\mathbf{y}^m$) on observed data ($\mathbf{y}^o$) and parameters (θ)

$$\mathbf{y}^m \mid \mathbf{y}^o, \theta \sim \mathcal{N}(\mu_{\mathbf{y}^m}, \mathbf{K}_{\mathbf{y}^m}^{-1})$$

- Using quarterly national GDP the length of regional GDP data can be quadrupled
- Constraints ensure the hierarchy is respected
 - ▶ Inter-temporal constraint: quarterly regional estimates sum up to observed annual value
 - ▶ Cross-sectional constraint: regional growth rates aggregate to the national growth rate

▶ Details

Forecast Error Variance-Covariance Matrix

Use the MF-obtained data $\mathbf{y}^m$ for the forecast reconciliation variance-covariance matrix instead of the true unobserved data

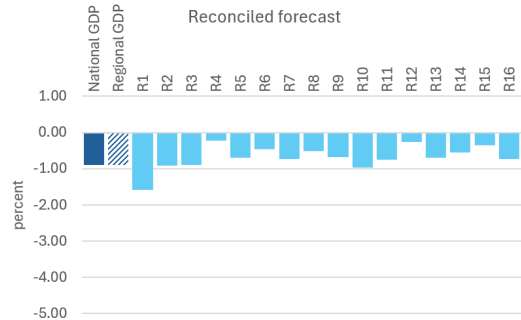
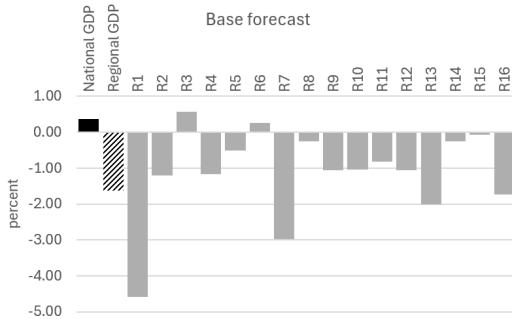
Proposition

Given $\mathbf{y}^m \xrightarrow{d} \mathbf{y}^{true}$, the MF-based forecast error variance-covariance matrix converges to the true variance covariance matrix $\text{var}(\hat{\mathbf{y}}_{T+h} - \mathbf{y}_{T+h}^m | \mathcal{I}_T) \xrightarrow{d} \text{var}(\hat{\mathbf{y}}_{T+h} - \mathbf{y}_{T+h}^{true} | \mathcal{I}_T)$

$$\mathbf{W}_h^m = \text{Var}(\hat{\mathbf{y}}_{T+h}^R - \mathbf{y}_{T+h}^{m,R})$$

$$\mathbf{W}_h^m \stackrel{\mathcal{D}}{=} \mathbf{W}_h$$

Application: Forecasting



Application: Forecasting German GDP reconciled vs unreconciled

- Real-time, out-of-sample forecasts for German GDP from 2005 to 2019
- Forecast accuracy relative to *base* forecasts from the same MF-VAR
- Standard Diebold-Mariano metrics

	DE	BW	BY	BE	BB	HB	HH	HE	MV
RMSFE	0.95**	0.93***	0.98	1	1.05	0.98**	0.95***	0.94***	1.03
CRPS	1.02	0.97	1.06	1.12	0.93**	1.02	1.04	1	0.98
	NI	NW	RP	SL	SN	ST	SH	TH	
RMSFE	0.96**	0.95**	0.96**	0.97*	1.06	0.98**	0.97**	0.99	
CRPS	0.98	1	0.96	0.92**	0.99	0.89***	1.09	0.94	

Reconciled IRFs

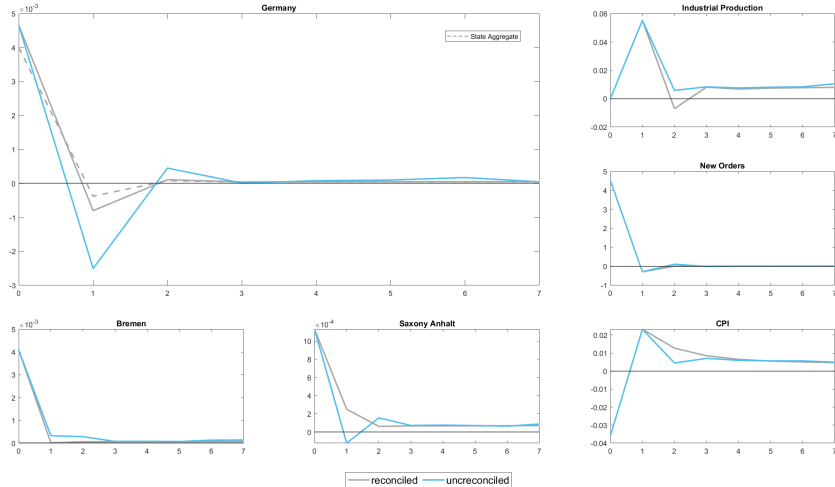
- IRFs may be calculated as the difference between a forecast with a shock and a forecast without a shock (Koop et al., 1996; Pesaran and Shin, 1998)
 - ▶ Linear model: any time period
 - ▶ Non-linear model: average over all time periods (Generalized impulse response functions)
 - ▶ Does not rely on identification scheme

$$IRF(h, \omega_{t-1}) = E(\hat{\mathbf{y}}_{t+h} | \epsilon_t, \omega_{t-1}) - E(\hat{\mathbf{y}}_{t+h} | \omega_{t-1})$$

- Reconciled impulse response functions

$$rIRF(h, \omega_{t-1}) = E(\tilde{\mathbf{y}}_{t+h} | \epsilon_t, \omega_{t-1}) - E(\tilde{\mathbf{y}}_{t+h} | \omega_{t-1})$$

Application: Structural analysis



Summary

- We extend forecast reconciliation methods to be used for macroeconomic hierarchical data (short data)
- State-of-the-art models can now be used for policy analysis
 - ▶ Reconciled forecasts with better properties (point and density)
 - ▶ Reconciled forecasts can be framed as conditional forecasts (implemented easily)
 - ▶ Reconciled impulse responses adhere to the hierarchical structure

Thank you!

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Mixed-Frequency Framework

Stacking a standard VAR(p) over T periods and rewriting it in matrix form

$$\mathbf{y}_t = \mathbf{b}_0 + \mathbf{B}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{B}_p \mathbf{y}_{t-p} + \varepsilon_t \quad \varepsilon_t \sim \mathcal{N}(\mathbf{0}_n, \sigma)$$

$$\mathbf{H}_B \mathbf{y} = \mathbf{c}_B + \varepsilon \quad \varepsilon \sim \mathcal{N}(\mathbf{0}_{Tn},)$$

Separate data into observed and unobserved

$$\mathbf{H}_B \mathbf{S}^o \mathbf{y}^o + \mathbf{H}_B \mathbf{S}^m \mathbf{y}^m = \mathbf{c}_B + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}_{Tn}, \text{Sigma}),$$

We can draw the missing observations conditional on the observed data and the constraints, $\mathbf{z}$, that ensure that the hierarchical system is fulfilled.

$$(\mathbf{y}^m | \mathbf{y}^o, \theta, \mathbf{z}) \sim \mathcal{N}(\hat{\mu}_{y^m}, \bar{\mathbf{K}}_{y^m}^{-1})$$

Conditional Forecasting

Conditional forecasting may be formulated as a set of linear restrictions r and Ω

$$\mathbf{R}\mathbf{y}_{T+1,T+h} \sim \mathcal{N}(\mathbf{r}, \Omega)$$

which can then be considered "observations" and the conditional forecasts may be drawn from

$$\mathbf{y}_{T+1,T+h}^m \mid \mathbf{y}_{T+1,T+h}^o, \boldsymbol{\theta}, \mathbf{r}, \sim \mathcal{N}\left(\boldsymbol{\mu}_{\mathbf{y}_{T+1,T+h}^m}, \mathbf{K}_{\mathbf{y}_{T+1,T+h}^m}^{-1}\right)$$

Reconciled Forecasts as Conditional Forecasts

Given the base now-/forecast $\hat{\mu}_{T+1:T+h}$, we can draw the reconciled forecast from

$$\tilde{\mathbf{y}}_{T+1:T+h} \sim \mathcal{N}(\Pi \hat{\mu}_{T+1:T+h}, \mathbf{0}_{r \times r})$$

where

$$\Pi = \begin{pmatrix} \mathbf{SP}_1\mathbf{R} & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \mathbf{SP}_h\mathbf{R} \end{pmatrix}$$

The variance-covariance matrix is set to zero to implement the reconciliation as a hard constraint.

Reconciled forecasts as conditional forecasts II

In case of an unconditional base forecast, i.e. no information in the forecast horizon, one can simply use

$$\tilde{\mathbf{y}}_{T+1:T+h} \sim \mathcal{N}(\Pi \mathbf{H}^{-1} \mathbf{c}, \mathbf{0}_{r \times r})$$